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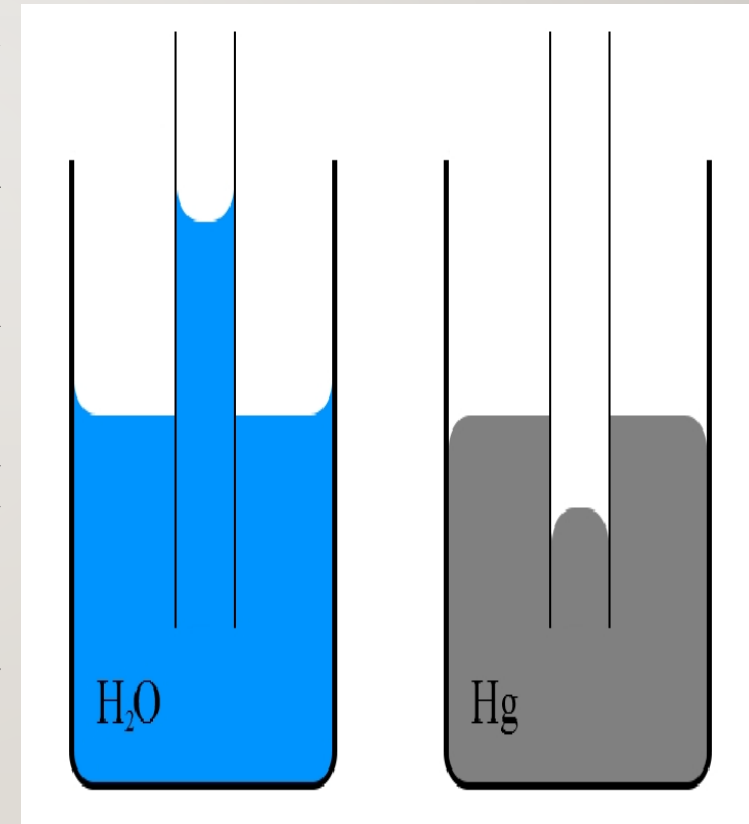


MECHANICAL PROPERTIES OF FLUIDS PART- 7(I)

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CAPILLARY ACTION

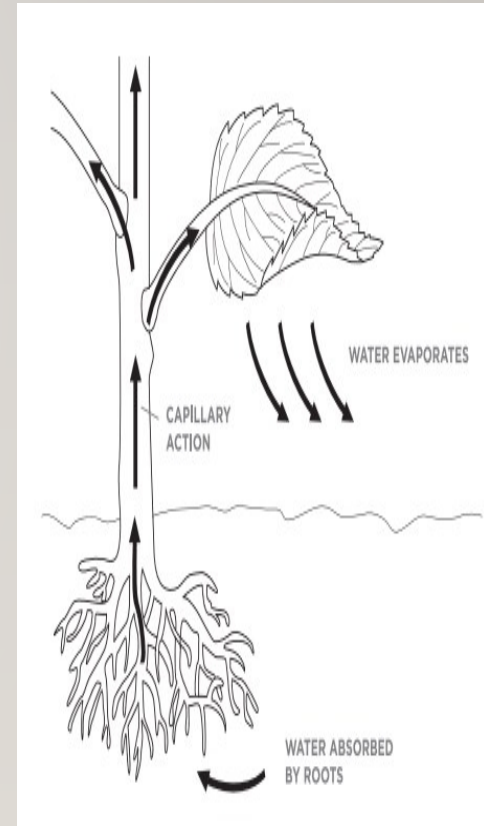
- A tube having a very fine bore (1mm) and open at both ends is called a capillary tube.
- If one end of a capillary tube is dipped in a liquid which partially or completely wets the surface of the capillary (like water in glass) the level of liquid in the capillary rises.
- On the other hand, if the capillary tube is dipped in a liquid which does not wet its surface (like mercury in glass) the level of liquid in the capillary drops.
- The phenomenon of rise or fall of a liquid inside a capillary tube when it is dipped in the liquid is called *capillarity*.



EXAMPLES :

Capillarity is in action when,

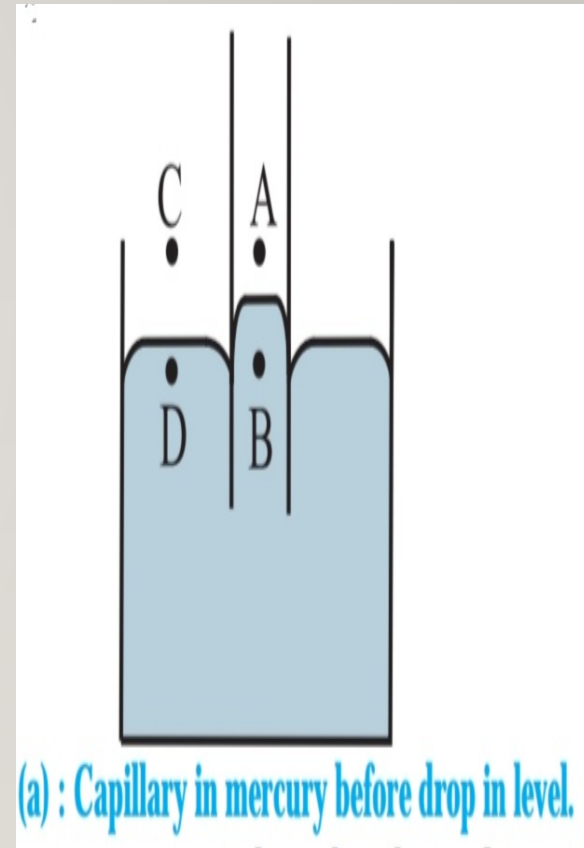
- Oil rises up the wick of a lamp.
- Cloth rag sucks water.
- Water rises up the crevices in rocks.
- Sap and water rise up to the top most leaves in a tree.
- Blotting paper absorbs ink.



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- When a capillary is dipped in a liquid, two effects can be observed,
 - a) The liquid level can rise in the capillary (water in a glass capillary), or
 - b) The liquid level can fall in the capillary (mercury in glass capillary).
 - Here we discuss a qualitative argument to explain the capillary fall.

CAPILLARY FALL:

- Consider a capillary tube dipped in a liquid which does not wet the surface, for example, in mercury.
- The shape of mercury meniscus in the capillary is upper convex.
- Consider the points A, B, C, and D such that, (see Fig. (a)).
 - i) Point A is just above the convex surface and inside the capillary.
 - ii) Point B is just below the convex surface inside the capillary.
 - iii) Point C is just above the plane surface outside the capillary.
 - iv) Point D is just below the plane surface and outside the capillary, and below the point C

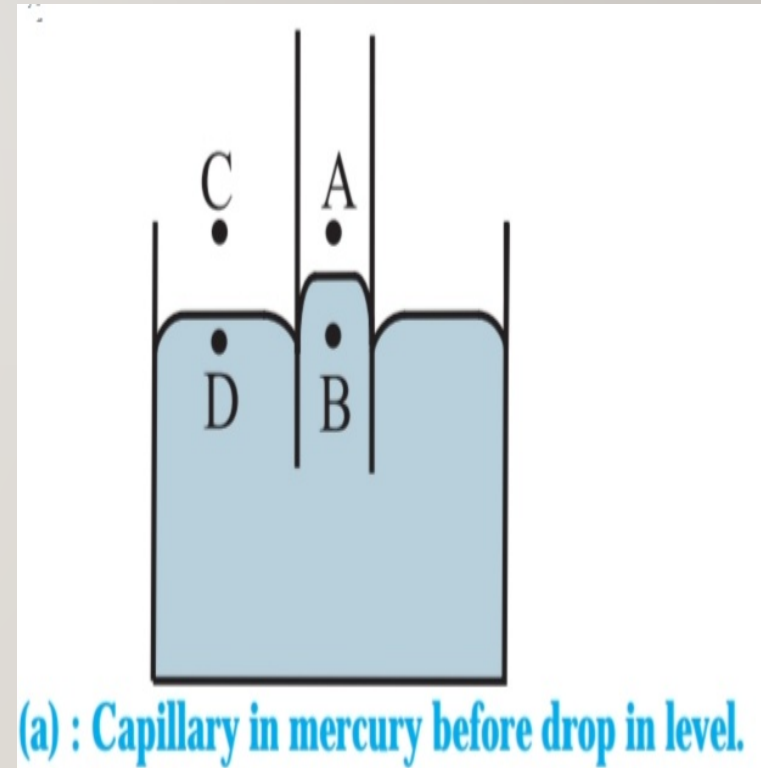


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- Let p_A , p_B , p_C , and p_D be the values of the pressures at the points A, B, C, and D respectively.
 - As discussed previously, the pressure on the concave side is always greater than that on the convex side.

$$P_B > P_A$$

- As the points A and C are at the same level, the pressure at both these points is the same, and it is the atmospheric pressure.

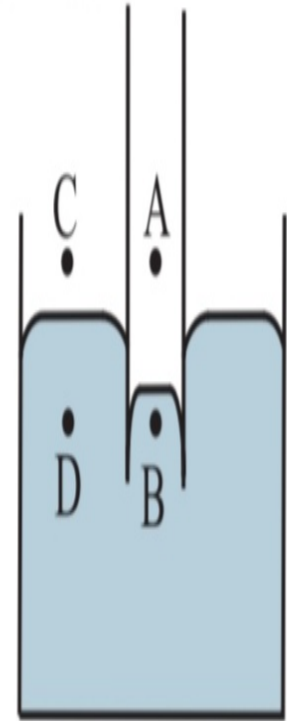
$$P_C = P_A$$



- Between the points C and D, the surface is plane.

$$P_C = P_A = P_D$$

- $P_B > P_D$, But the points B and D are at the same horizontal level.
- Thus, in order to maintain the same pressure, the mercury in the capillary rushes out of the capillary.
- Because of this, there is a drop in the level of mercury inside the capillary as shown in Fig.



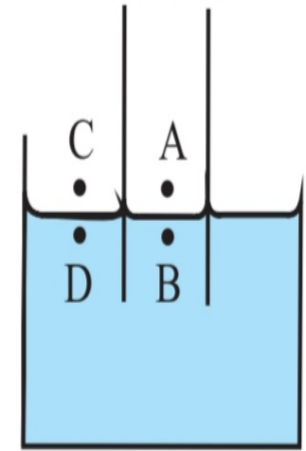
(b): Capillary in mercury, drop in level.

CAPILLARY RISE: EXPRESSION FOR CAPILLARY RISE OR FALL: METHOD (I): USING PRESSURE DIFFERENCE

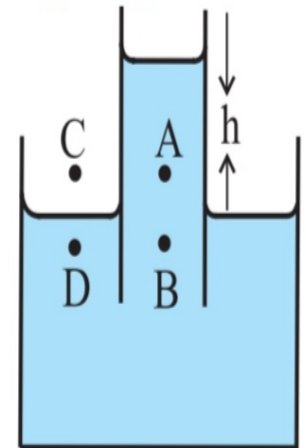
- Refer to Fig. (a) and Fig. (b) and explain the rise of a liquid inside a capillary.
- The pressure due to the liquid (water) column of height h must be equal to the pressure difference $2T/R$ due to the concavity.

$$h\rho g = \frac{2T}{R}$$

where, ρ is the density of the liquid and g is acceleration due to gravity.



(a): Capillary just immersed in water.



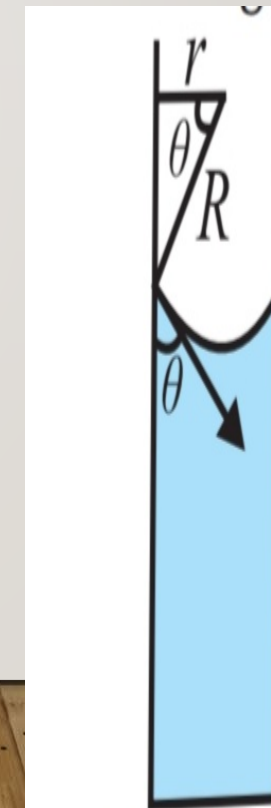
(b): Capillary in water after rise in level.

- Let r be the radius of the capillary tube and θ be the angle of contact of the liquid as shown in Fig. (a).

Then radius of curvature R of the meniscus
is given by $R = \frac{r}{\cos \theta}$

$$\therefore h\rho g = \frac{2T\cos\theta}{r}$$

$$\therefore h = \frac{2T\cos\theta}{r\rho g}$$



Forces acting at
the point of contact.

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- The above equation gives the expression for capillary rise (or fall) for a liquid.
 - Narrower the tube, the greater is the height to which the liquid rises (or falls).
 - If the capillary tube is held vertical in a liquid that has a convex meniscus, then the angle of contact θ is obtuse.
 - Therefore, $\cos\theta$ is negative and so is h .
 - This means that the liquid will suffer capillary fall or depression.

(METHOD II): USING FORCES:

- Rise of water inside a capillary is against gravity.
- Hence, weight of the liquid column must be equal and opposite to the proper component of force due to surface tension at the point of contact.
- The length of liquid in contact inside the capillary is the circumference $2 \pi r$.
- Thus, the force due to surface tension is given by,

$$f_T = (\text{surface tension}) \times (\text{length in contact}) \\ = T \times 2\pi r$$

Direction of this force is along the tangent, as shown in the Fig. 2.25 (b).

Vertical component of this force is

$$(f_T)_v = T \times 2\pi r \times \cos \theta \quad \text{--- (2.29)}$$

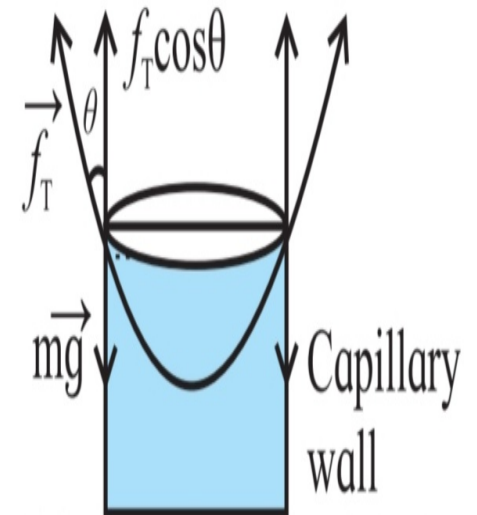


Fig. 2.25 (b): Forces acting on liquid inside a capillary.

Ignoring the liquid in the concave meniscus, the volume of the liquid in the capillary rise is $V = \pi r^2 h$.

$\therefore$ Mass of the liquid in the capillary rise,

$$m = \pi r^2 h \rho$$

∴ Weight of the liquid in the capillary (rise or fall), $w = \pi r^2 h \rho g$ --- (2.30)

This must be equal and opposite to the vertical component of the force due to surface tension. Thus, equating right sides of equations (2.29) and (2.30), we get,

$$\pi r^2 h \rho g = T \times 2\pi r \times \cos \theta$$

$$\therefore h = \frac{2T \cos \theta}{r \rho g}$$

In terms of capillary rise, the expression for surface tension is,

$$T = \frac{r h \rho g}{2 \cos \theta} \quad \text{--- (2.31)}$$

The same expression is also valid for capillary fall discussed earlier.